(MSE) for regression tasks and CrossEntropy Loss for classification tasks.

- **Backpropagation**: Backpropagation is an optimization technique used to update the weights and biases of a neural network. It involves:
    1. Calculating the loss using the loss function.
    2. Propagating the error backward through the network.
    3. Updating weights and biases using algorithms like gradient descent, which minimizes the loss function by iteratively adjusting the parameters.

Applications of Neural Networks

Neural networks have revolutionized numerous domains by enabling machines to perform tasks once considered exclusive to humans. Some notable applications include:

- **Image Recognition**: Identifying objects, faces, or scenes in images, with applications in security and social media.
- **Natural Language Processing (NLP)**: Understanding and generating human language for chatbots, translation systems, and sentiment analysis.
- **Medical Diagnosis**: Assisting in detecting diseases from medical images like X-rays or MRIs.
- **Financial Forecasting**: Predicting stock prices and assessing credit risk.
- **Autonomous Vehicles**: Enabling self-driving cars to perceive their environment and make decisions.

The Future of Neural Networks

As computational power continues to grow and access to large datasets becomes more prevalent, the capabilities of neural networks are expected to expand further. Innovations such as transfer learning, neural architecture